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Clinical Applications of Wearable Sensor-Based Gait Analysis in Athletes: A Systematic Review for Injury Prevention and Rehabilitation

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Abstract

Background: Gait analysis is fundamental for optimizing athletic performance and mitigating injury risks in sports medicine. While traditional laboratory assessments offer insights, the emergence of wearable sensor technology provides novel opportunities for objective, real-time gait evaluation in ecologically valid settings. This systematic review aims to (1) evaluate the clinical efficacy of wearable sensor–based gait analysis for injury prevention in athletes, (2) examine its role in rehabilitation monitoring and clinical decision-making, and (3) identify methodological limitations and future research priorities required for standardized clinical integration.

Methods: Adhering to PRISMA 2020 guidelines, a systematic search was conducted across MEDLINE, Web of Science, Scopus, and Embase from inception to May 31st, 2025. Studies investigating wearable sensor-based gait analysis in athletic populations for clinical applications (injury prevention, diagnosis, rehabilitation, or performance-related decision-making) were included. Data on study design, participant characteristics, sports, clinical conditions, sensor types, and gait parameters were extracted.

Results: From 2678 initial records, 22 studies (14 comparative, 8 non-comparative observational) met the inclusion criteria, encompassing 1040 participants: 841 athletes (weighted mean age: 25.7 years) and 199 healthy controls (weighted mean age: 31.4 years). Running (7 studies) and anterior cruciate ligament (ACL) injuries (6 studies) were the most frequently investigated sport and clinical condition, respectively. Inertial Measurement Units (IMUs) were the predominant sensor technology (59.1%), primarily assessing parameters like tibial shock, vertical ground reaction force, and spatio-temporal variables. Wearable sensors demonstrated emerging evidence of utility in identifying biomechanical alterations indicative of injury risk, monitoring fatigue, and guiding rehabilitation protocols. Key challenges identified include heterogeneity in methodologies, data accuracy concerns, and the need for standardized reporting.

Conclusion: Wearable sensor–based gait analysis shows emerging evidence of providing objective biomechanical insights that may support injury risk assessment and rehabilitation monitoring in athletic populations. Evidence from moderate-quality observational studies indicates that simple, well-placed sensors may yield clinically meaningful information on loading and spatiotemporal control, complementing standard examination and decision-making in sports medicine. In practice, tibial accelerometers may be particularly useful for monitoring impact loads relevant to bone stress injuries, while trunk-mounted inertial measurement units show promise for

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providing spatiotemporal parameters to guide rehabilitation and return-to-play decisions. These applications are feasible in team and clinical environments, require minimal setup, and provide repeatable metrics that can directly inform athlete management. Broader adoption, however, will depend on improving methodological consistency, reporting standards, and integration with other clinical data sources. Future work using adequately powered samples and standardized sensor protocols will be essential to strengthen the evidence base and support clinical implementation.

Keywords: Gait analysis, wearable sensors, athletes, sports medicine, injury prevention, rehabilitation, biomechanics, performance enhancement.

Introduction

The analysis of movement in sports, with a particular focus on gait assessment, plays a critical role in enhancing athletic performance and mitigating injury risks (1). In recent years, the systematic study of athletes' biomechanics has advanced considerably, equipping medical professionals with improved tools to identify potential injury predispositions and refine training regimens (2, 3). With sports injuries impacting millions of athletes annually, the capacity for precise measurement and analysis of movement patterns is fundamental for informing evidence-based clinical decisions in sports medicine (4).

Recent advances in wearable sensor technology have transformed traditional gait analysis methods. These sensors enable continuous monitoring of athletes during real-world training and competition, moving beyond the limitations of laboratory settings. This technology measures key movement parameters such as stride length, speed, and limb coordination. Such detailed biomechanical assessment aids in identifying potential injury risks (5-7). Wearable sensors also allow for a more personalized approach to training and rehabilitation. By providing continuous monitoring of an athlete's biomechanical data in real-life situations, this technology thereby facilitates individualized, data-driven athletic care (6, 8).

The clinical utility of wearable gait analysis technologies in sports medicine is substantial, particularly for injury prevention via early detection of biomechanical abnormalities and the subsequent implementation of targeted interventions (7, 9). These systems provide quantitative measures that are crucial for monitoring rehabilitation progress and guiding evidence-based decisions concerning return-to-play protocols (10-12). Moreover, the detailed biomechanical data acquired facilitates precise assessment of movement patterns, thereby enabling the development of individualized rehabilitation programs tailored to address specific deficits (1).

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Despite these significant advantages, the integration of wearable sensor technology into routine clinical practice and athletic training faces several critical challenges. Prominent among these are concerns surrounding data accuracy and reliability, the necessity for robust privacy protection mechanisms, effective strategies for comprehensive data management, and the crucial step of translating complex biomechanical data into actionable clinical insights (13). Furthermore, the establishment of standardized protocols is crucial to ensure methodological consistency across studies, thereby facilitating meaningful comparative analyses and strengthening the generalizability of findings (1, 14, 15). Addressing these multifaceted challenges is vital to fully realize and maximize the clinical utility of wearable sensor technology in sports medicine.

While previous systematic reviews have explored wearable sensor technology in gait analysis, specific limitations in their scope and timeliness underscore the need for the present research. For instance, Mason et al. (16) focused solely on running gait with inertial measurement units (IMUs) in young, recreational runners, limiting the generalizability of their findings to other athletic populations and sporting activities. Saboor et al. (17) examined wearable sensors and machine learning; however, their review, with a search likely concluding in early 2020, does not capture the most recent advancements in this rapidly evolving field. Similarly, Hutabarat et al. (18) reviewed quantitative gait analysis using wearable sensors, but their inclusion of studies not specifically targeting clinical applications or using varied, non-standardized devices dilutes the practical utility for sports medicine. Together, these reviews highlight a clear gap for an updated synthesis that focuses specifically on clinical applications in athletic populations.

In this work, clinical applications are operationally defined as uses of wearable gait data that directly inform clinical decisions, specifically including: (a) injury risk stratification and prevention strategies, (b) rehabilitation progression monitoring, (c) return-to-play readiness assessment, and (d) training load optimization based on biomechanical parameters.

The focus on athletes is essential because their biomechanical demands, exposure to repetitive high-intensity loading, and return-to-play expectations differ markedly from those of general or clinical populations. Wearable gait assessment in sports settings requires interpretation within the context of sport-specific movement patterns, fatigue, and performance objectives that shape both injury mechanisms and rehabilitation progress. Concentrating on athletic cohorts therefore provides insights directly relevant to sports medicine practice, which may not be transferable from non-athletic populations. While this focus enhances clinical relevance for sports medicine, it may limit generalizability to non-athletic or sedentary populations.

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The field of wearable sensor technology is advancing rapidly, with recent integration of machine learning enabling accurate estimation of whole-body kinetics and stride kinematics across varied running speeds (19). Considering this dynamic landscape and the increasing adoption of these technologies in sports medicine (20), a clear need exists to consolidate current evidence. This review aims to: (1) evaluate the clinical efficacy of wearable sensor–based gait analysis for injury prevention in athletes, (2) examine its role in rehabilitation monitoring and clinical decision-making, and (3) identify methodological limitations and future research priorities required for standardized clinical integration.

Methods

This systematic review was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines (21)and was prospectively registered in PROSPERO (CRD42024555901). All procedures, including search strategy, eligibility criteria, and synthesis plan, followed the registered protocol without deviations. The literature search was performed across MEDLINE (via PubMed), Web of Science, Scopus, and Embase, covering the period from database inception to May 31st, 2025. The search strategy included keywords and Medical Subject Headings (MeSH) terms related to “gait analysis” and “wearable sensors,” applied to the title and abstract fields of each database.

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Eligibility Criteria

Studies were eligible for inclusion if they examined the clinical applications of wearable sensor-based gait analysis in athletes, defined as individuals who engage in structured and regular training for competitive sports at professional, semi-professional, or recreational levels. "Clinical applications" were defined as the use of wearable gait data to inform injury prevention strategies, rehabilitation protocols, diagnosis, monitoring, or performance-related clinical decision-making, such as modifying training load based on gait metrics, designing a rehabilitation protocol or determining readiness for return-to-play. Studies were included regardless of language or study design, provided they included at least five participants and reported relevant data. Studies with fewer than five participants were excluded to ensure a minimum sample size for meaningful data reporting and to minimize the risk of spurious findings from very small cohorts. Only peer-reviewed full-text articles were considered for inclusion; grey literature sources, including conference abstracts, dissertations, preprints, and non–peer-reviewed reports, were excluded to ensure inclusion of studies with complete methodological and peer-reviewed reporting. This exclusion is also

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noted in the Discussion section to clarify its rationale and potential influence on publication bias.

Studies were excluded for methodological shortcomings if they: (a) did not clearly describe sensor type, placement, or sampling frequency; (b) lacked adequate description of data processing algorithms; (c) provided insufficient participant characteristic details; or (d) did not report validity/reliability metrics for novel applications.

Study selection

Following the removal of duplicates using EndNote (version 21), two reviewers (A.A. and M.M.T.) independently screened the titles and abstracts of the retrieved records to identify potentially relevant studies. Any disagreements were resolved through discussion and consensus, with the involvement of a third reviewer (V.S.), the senior author, when necessary. Full-text articles of the studies deemed potentially eligible were then independently assessed for final inclusion. Inter-rater agreement for full-text inclusion decisions was excellent ($\kappa = 0.86$), indicating high consistency between reviewers. This kappa value applies exclusively to the full-text screening phase; title and abstract screening were conducted independently, and discrepancies were resolved by consensus without formal statistical calculation.

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The following search strategy was developed and adapted for each database:

PubMed:

(("Gait Analysis"[Mesh] OR "gait analysis" OR "gait biomechanics" OR "gait assessment" OR "spatiotemporal gait parameters" OR "stride length" OR "gait characteristics" OR "gait patterns" OR "kinematic gait analysis" OR "biomechanical gait analysis") AND ("Wearable Electronic Devices"[Mesh] OR "wearable sensors" OR "inertial measurement units" OR IMU OR accelerometer OR gyroscope OR magnetometer OR "insole sensors" OR "electromyography sensors" OR "motion capture" OR "wearable technology") AND ("Sports"[Mesh] OR athlete OR athletes OR "sports performance" OR "competitive sports") AND ("Rehabilitation"[Mesh] OR rehabilitation OR "injury prevention" OR "return to play" OR "sports medicine" OR "performance monitoring" OR "clinical application" OR diagnosis OR monitoring))

Embase: A similar search strategy to PubMed was employed, adapting MeSH terms to Embase’s controlled vocabulary (Emtree terms) where appropriate.

Scopus:

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TITLE-ABS-KEY(("gait analysis" OR "gait biomechanics" OR "gait assessment" OR "spatiotemporal gait parameters" OR "stride length" OR "gait characteristics" OR "gait patterns" OR "biomechanical gait analysis" OR "kinematic gait analysis") AND ("wearable sensors" OR "wearable devices" OR "inertial measurement units" OR IMU OR accelerometer OR gyroscope OR magnetometer OR "insole sensors" OR "electromyography sensors" OR "motion capture" OR "wearable technology") AND (athlete OR athletes OR "sports performance" OR "competitive sports") AND ("rehabilitation" OR "injury prevention" OR "return to play" OR "sports medicine" OR "performance monitoring" OR "clinical application" OR diagnosis OR monitoring))

Web of Science:

TITLE-ABS-KEY(("gait analysis" OR "gait biomechanics" OR "gait assessment" OR "spatiotemporal gait parameters" OR "stride length" OR "gait characteristics" OR "gait patterns" OR "biomechanical gait analysis" OR "kinematic gait analysis") AND ("wearable sensors" OR "wearable devices" OR "inertial measurement units" OR IMU OR accelerometer OR gyroscope OR magnetometer OR "insole sensors" OR "electromyography sensors" OR "wearable technology") AND (athlete OR athletes OR "sports performance" OR "competitive sports") AND ("rehabilitation" OR "injury prevention" OR "return to play" OR "sports medicine" OR "performance monitoring" OR "clinical application" OR diagnosis OR monitoring))

Complete database-specific search strings for MEDLINE, Web of Science, Scopus, and Embase are provided in Supplementary Table S2.

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Data collection

Two reviewers (A.A. and M.M.T.) independently extracted the following data from each included study using a standardized data extraction form: study design (e.g., comparative, observational), participant characteristics (number of participants, sex, athlete type [professional/recreational]), clinical condition (if applicable), sport type, gait analysis environment (e.g., laboratory, field), wearable sensor type(s), sensor placement, and specific gait parameters measured. Any disagreements during the data extraction phase were resolved through discussion and consensus. If consensus was not reached, a third senior reviewer (V.S.) was consulted to make a final decision. All extracted data were cross-verified by the third reviewer (V.S.) to ensure completeness and accuracy. Data extraction was independently verified for all 22 included studies, with no discrepancies identified.

Statistical analysis

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Due to the heterogeneity in study designs, sensor technologies, clinical conditions, and reported gait parameters, a meta-analysis was not feasible. Therefore, a narrative synthesis approach was employed. Findings were categorized and summarized thematically based on clinical condition (e.g., ACL injuries, fatigue, hip/groin pain), sensor type and placement (e.g., IMU, sEMG, accelerometers), gait parameters measured (e.g., tibial shock, muscle activation, vGRF), and sport or population characteristics. When studies presented contradictory or inconsistent findings for comparable clinical conditions or sensor types, discrepancies were examined systematically by considering differences in sensor placement, data processing algorithms and derived metrics, sample characteristics, and overall methodological quality. This structured approach enhanced transparency and consistency in interpreting heterogeneous evidence.

This thematic approach facilitated comparison of similar studies and identification of consistent patterns in clinical applications. No formal sensitivity analyses were performed because of substantial heterogeneity in study designs, sensor technologies, placement protocols, and outcome definitions, which precluded defensible subgroup or stratified analyses.

Quality assessment

Two independent reviewers assessed the methodological quality of included studies using the Methodological Index for Non-Randomized Studies (MINORS) criteria (22). The MINORS tool assigns scores from 0-2 for each item (not reported, reported but inadequate, or reported and adequate), with maximum possible scores of 16 for non-comparative studies (8 items) and 24 for comparative studies (12 items). Higher scores indicate better methodological quality. Disagreements were resolved through discussion or by consulting a third reviewer when necessary. MINORS scores were interpreted relative to the score distribution across included studies; no standardized cutoffs were applied. (Table 1)

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Study	Minor score
Ihara et al. 2008	17/24
Sakurai et al. 2010	19/24
Patterson et al. 2014	14/24
Cherati et al. 2016	9/16
Gilgen-Ammann et al. 2016	21/24
Hansen et al. 2017	16/24
Dan et al. 2019	19/24
Mansourizadeh et al. 2019	16/24
Clermont et al. 2020	12/16
Sadeghi et al. 2020	17/24
Lawrenson et al. 2021	17 /24
Tajdini et al. 2021	9/16
De Jong et al. 2022	11/16
Oldham et al. 2022	18 /24
Riedl et al. 2022	11/16
Hunzinger et al. 2023	18/24
Popp et al. 2023	17/24
Versloot et al. 2023	19/24
Zandbergen et al. 2023	9/16
Fuller et al. 2024	11/16
Han R et al. 2024	16/24
Thapa RK et al. 2024	13/16

Table 1. Quality assessment of the included studies based on the MINORS scoring criteria

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Results

Study Selection

The database searches identified 2678 records. After removing 1121 duplicates, 1557 records underwent title and abstract screening, resulting in 101 articles eligible for full-text review. Of these, 79 articles were excluded primarily due to (specific reasons: e.g., irrelevant populations (n=26), absence of gait analysis (n=18), non-wearable sensor technology (n=15), and insufficient data reporting (n=20)). The systematic review ultimately included 22 studies that met all inclusion criteria (Figure 1).

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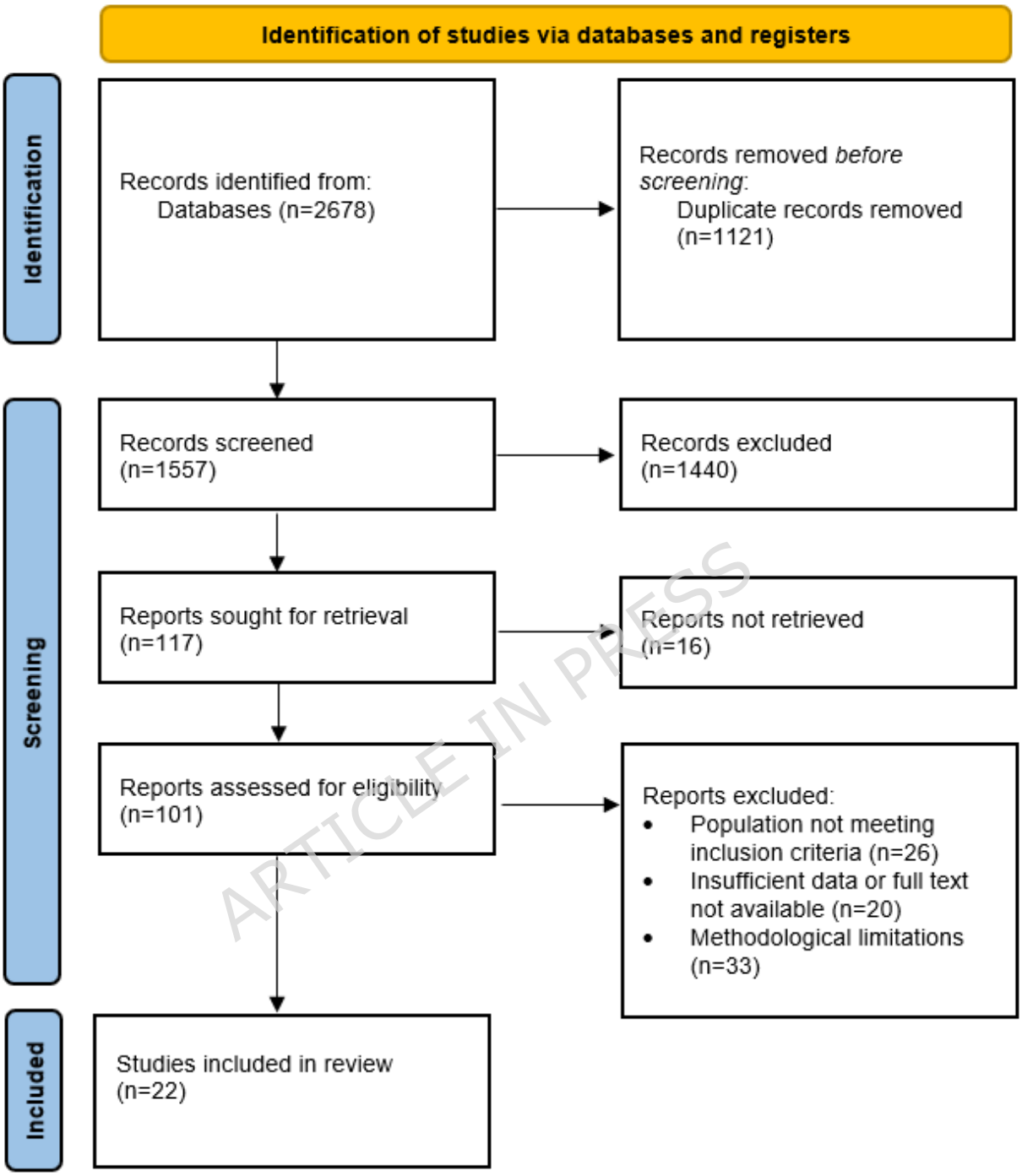

Figure 1. PRISMA flow diagram

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Study Characteristics

The 22 included studies comprised 13 (59.1%) comparative and 9 (40.9%) non-comparative designs, encompassing a total of 1040 participants. Of these, 841 were athletes (with or without specific clinical conditions) with a weighted mean age of 25.7 years, and 199 were healthy controls with a weighted mean age of 31.4 years. All studies were observational; no interventional studies met the inclusion criteria. The included studies were published in English between 2008 and 2024, with 14 (63.6%) published within the last five years (2020-2024). Further details on study characteristics are presented in Table 2.

Methodological Quality of Included Studies

The methodological quality of the included studies was assessed using the MINORS criteria, with scores ranging from 0–24 for comparative studies and 0–16 for non-comparative studies. We considered scores ≥12/16 for non-comparative and ≥20/24 for comparative studies as indicating relatively higher methodological rigor, representing the upper 50th percentile. These thresholds were determined by the review team based on score distribution and do not represent validated cutoffs from MINORS developers.

For the 14 comparative studies, scores ranged from 14 to 21, with a median of 17. Only one study (7.1%) achieved a score ≥20, while the majority fell within a moderate range. Common methodological strengths included clearly defined research objectives, appropriate participant selection, and use of reliable outcome measures. Recurring limitations were the absence of standardized wearable sensor protocols and placement reporting, incomplete follow-up data (e.g., attrition rates), and variability in statistical analyses.

For the 8 non-comparative studies, scores ranged from 9 to 13, with a median of 11. Two studies (25%) met the threshold for relatively higher methodological rigor (≥12/16). Methodological weaknesses were particularly evident in this group, including small sample sizes, lack of control groups, potential selection bias, and incomplete data reporting.

Among the 22 included studies, 7 (31.8%) enrolled fewer than 20 participants, 1 (4.5%) did not report sensor sampling frequency, 6 (27.3%) did not specify data-filtering algorithms, and 8 (36.4%) included no control group. These proportions were derived from the standardized extraction table for all included studies.

Across both comparative and non-comparative designs, the assessment revealed considerable variability in methodology, particularly in sensor types, placement, data collection, and outcome measures. Taken together, these findings indicate that while a minority of studies demonstrated relatively higher methodological rigor by MINORS standards, most provided only moderate-quality evidence, underscoring the need for caution in clinical translation. These methodological shortcomings may influence clinical

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interpretation. For example, small sample sizes and lack of control groups reduce confidence in the generalizability of findings and data processing limits reproducibility of gait parameters in clinical practice, and heterogeneity in outcome measures hinders the development of standardized recommendations for injury prevention and rehabilitation.

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Study (year)	Participants, n (M:F)	Age, mean ± SD	Sensor configuration (type, placement, sampling frequency)	Sport / Group or Condition	Key gait parameters	Key findings
Tajdini 2021 (23)	28:0	23.7 ± 2.1	Force sensor; bilateral feet; 500 Hz + sEMG; rectus femoris and biceps femoris; 1000 Hz	ACLR (isolated)	vGRF; rectus femoris and biceps femoris activity (contact phase)	Fear of reinjury positively correlated with asymmetry in 2nd peak vGRF and rectus femoris/biceps femoris activity during contact phase.
Versloot 2023 (24)	125:0	25 ± 12	Triaxial accelerometer (Activ8); right upper thigh; 500 Hz	Hemophilia, sports injuries	Time spent in dynamic activities	Endurance and physical activity did not predict increased risk of sports injuries or SIBs in patients with hemophilia.
Sadeghi 2020 (25)	40:0	ACLD 25.6 ± 5.0; Healthy 26.2 ± 3.0	Triaxial accelerometer; tibial tuberosity; 100 Hz	Comparative: ruptured ACL/ Healthy individuals	Loading factor trajectories	Gait differences in ACL-deficient individuals primarily at end of mid-swing, beginning of terminal swing (vertical axis), and stance phase (anterior-posterior axis).
Sakurai 2010 (26)	25:0	Painful knee 12.0 ± 1.4; Control 12.8 ± 1.7	Triaxial piezoelectric accelerometers; lateral malleolus and head of fibula; 100 Hz	Chronic sports knee injury / Healthy controls	Peak acceleration values	Significantly increased peak acceleration in painful knee group, indicating increased stress in all directions.
Ihara 2008 (27)	18:6	ACLD 23.1 ± 7.8; Control 21.8 ± 0.6	Triaxial piezoelectric accelerometers; lateral side of both knees; 1000 Hz	ACLR patients /Healthy athletes	Stepping power; stepping reaction (mediolateral and vertical acceleration amplitudes)	Healthy: stronger stepping power on tilted side, faster reaction on opposing side. ACLR: patterns reversed, indicating impaired postural control.

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Lawrenson 2021 (28)	61:0	Hip pain 29 ± 7; Control 29 ± 7	sEMG; gluteus maximus, gluteus medius, tensor fasciae latae, Adductor longus, Rectus femoris; 2000 Hz	Football players with hip-related pain / Asymptomatic controls	Hip muscle activity (rectus femoris, gluteus maximus)	Differences in rectus femoris and gluteus maximus activity identified. Symptom severity may influence muscle activity.
Mansourizadeh 2019 (29)	32:0	LSGP: 25.56 ± 2.36; Control: 25.25 ± 2.11	sEMG + IMU; internal oblique, transversus abdominis, multifidus, adductor longus, and gluteus medius; IMU at sacrum (L5–S1); 1500 Hz (sEMG)	Insidious onset groin pain (≥ 6 weeks) / Healthy controls	AEMG; co-contraction ratio (thigh and thoracic muscles)	Less muscular activity and co-contraction during turning in groin pain group. Reversed during stance phase following rotation.
Hansen 2017 (30)	37:0	Control: 35.4 ± 7.8 years; HS Graft: 26 ± 3.84; BTB Graft: 27 ± 7.69	sEMG + accelerometer; soleus, medial/lateral gastrocnemius, and medial/lateral hamstrings; 2000 Hz (EMG)	ACLR patients/ Healthy individuals	Heel contact timing; muscle activation patterns during running	In control participants, different activation patterns were seen among different body weight conditions. No differences were observed in bone-to-bone ACLR participants, but differences were noted in hamstring ACLR comparing certain body weight conditions.
Cherati 2016 (31)	10:13	16.10 ± 1.26	sEMG; bilateral vastus medialis and vastus lateralis; electrogoniometer at lateral femoral condyle; 1000 Hz	Anterior Knee pain	Vastus medialis and lateralis muscle activity	Decreased muscle activity voltage in all phases of Dachi after six weeks of training, significant reduction at start and finish.

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Han R 2024 (32)	46:38	30.72 ± 8.91	IMU; bilateral thighs, waist, and ankles, sEMG; quadriceps, hamstring, and calf muscles; NR	Healthy and injured recreational to competitive runners	Stride length, cadence, joint angles (knee, ankle dorsiflexion, hip flexion), GRF, muscle activation, foot pressure distribution, vertical displacement.	Developed an ML model to predict running-related injuries (88.37% accuracy). Identified GRF, foot pressure, and stride length as key factors. Showed biomechanical differences between male/female runners and pre/post-injury. Higher cadence was negatively associated with injury severity.
Thapa RK 2024 (33)	15:15	Male: 21.0 ± 2.5; Female: 21.5 ± 2.1	IMU; L5 lumbar spine; 100 Hz	Healthy collegiate athletes (basketball, handball)	Speed, Cadence, Stance phase, Swing phase, First double support phase, Single support phase, Stride length, Step length.	Good to excellent intra-session reliability (ICC 0.807–0.978) and moderate to excellent inter-session reliability (ICC 0.683–0.931) for most gait variables. IMU is a reliable tool for gait pattern monitoring.
Fuller, J.T. 2024 (34)	5:4	37.8 ± 4.0	IMU; distal medial tibia, 504 Hz	Healthy recreational runners (marathon or half-marathon training)	DFA α1of stride interval, running economy, velocity at VO2max, subjective markers (fatigue, soreness, stress, sleep quality)	DFA α1 was sensitive to changes in fatigue over the training period; decreased with increased training load and increased during recovery. Significantly correlated with running economy changes.
Hunzinger 2023 (35)	53:60	34.88 ± 11.80	IMU; both feet and L5 vertebra; 128 Hz	Contact/collision sport history / Non-contact sport history	Double support time (% of gait cycle), gait speed (m/s), and stride length (m)	History of contact/collision sport did not negatively affect or predict neurobehavioral function.
Zandbergen 2023 (36)	4:3	36 ± 11	IMU; sternum, pelvis, lateral thighs, proximal tibiae, and midfoot bilaterally; 240 Hz	Healthy recreational runners with fatigue	Speed; stride frequency; peak tibial acceleration	Speed and stride frequency influence interpretation of changes in mechanical quantities (fatigue).
Popp 2023 (37)	0:45	26.7 ± 4.7	IMU; bilateral distal anteromedi	History of BSI / No BSI	VALR; VILR; vertical stiffness; tibial shock	Tibial shock increased from fresh to exerted state in multiple BSI group, but not in controls. No difference in load rates between groups.

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			al tibia; 1000 Hz			
Oldham 2022 (38)	168:0	19.2 ± 1.3	IMU; bilateral feet, L5; 128 Hz	Student-athletes from collision sports	Gait speed; stride length	Greater exposure to repetitive head impacts not detrimental to dynamic postural control.
De Jong 2022 (39)	9:13	Male: 20 ± 1.0; F: 20 ± 1.0	IMU; mounted bilaterally on shoes; 200 Hz	Healthy athletes with fatigue	Stride length; contact time; pronation	Stride length, impact, pace, and contact time significantly associated with session RPE.
Riedl 2022 (40)	5:4	Male: 31.6 ± 9.2; Female: 25.5 ± 9.1	IMU; pelvis, thighs, shanks, and feet; 200 Hz	Grade 4 sacral stress fracture	vGRF	No clear orthopedic or biomechanical risk factors (lateral asymmetry) for sacral stress fracture identified.
Clermont 2020 (41)	7:10	Male: 47.71 ± 15.32; Female: 34.20 ± 5.67	IMU; L3–L5; 200 Hz	Healthy runners with fatigue	Speed; cadence; step time; stride time; peak accelerations	Single tri-axial accelerometer with advanced signal processing can identify atypical motion due to fatigue, recovery, and pain.
Dan 2019 (42)	64:38	ACL-R: 33.84 ± 10.05; Normal: 25.93 ± 9.71; Elite: 22.80 ± 3.64	IMU; tibia; 100 Hz	ACLR / Non-athletic controls / Elite athletes	Coronal lower limb alignment	Differences in coronal alignment during single-leg activities in ACLR patients.
Gilgen-Ammann 2016 (43)	7:5	35.7 ± 10.1	IMU; mounted bilaterally on shoes; 1000 Hz	History of lower limb injury	Ground contact time (bilateral, for asymmetry detection)	Well-trained athletes with injury history showed greater gait asymmetry compared to those without. High-intensity runs over short distances recommended for asymmetry detection.
Patterson 2014 (44)	0:31	ACL-R: 23.7 ± 3.12 years; Controls: 20.8 ± 1.17	IMU; tibia; 100 Hz	ACLR (BPTB autograft) / ACLR (hamstring autograft)	Gait velocity	ACLR group showed kinematic and kinetic deviations from controls, but no temporal or spatial deviations.

Table 2. Summary of studies included in the systematic review

Values are mean ± SD unless stated. NR, not reported. ACL, anterior cruciate ligament; ACLD, ACL-deficient; ACLR, ACL reconstruction; AEMG, average electromyography amplitude; BPTB, bone-patellar tendon-bone; BSI, bone stress injury; CV, coefficient of variation; DFA α1, detrended fluctuation analysis alpha 1; EMG, electromyography; GRF, ground reaction force; HS, hamstring tendon autograft; ICC, intraclass correlation coefficient; IMU, inertial measurement unit; ML, machine learning; ROM, range of motion; RPE, rate of perceived exertion; SD, standard deviation; SEM, standard error of measurement; sEMG, surface electromyography; SIB, sports-induced bleed; VALR,

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vertical average loading rate; VILR, vertical instantaneous loading rate; vGRF, vertical ground reaction force.

Overview of Sports and Clinical Conditions Investigated

Sports Investigated

Running was the most frequently studied sport (seven studies), followed by rugby (three studies). One study included both rugby and hockey players. Other sports represented across the included studies were handball, karate, soccer, cross-country, basketball, and Australian football. Five studies did not specify a particular sport.

Key Clinical Conditions Addressed

ACL injuries were the most commonly investigated clinical condition, representing 27.3% (6/22) of the studies. Hip and groin pain were the focus of 9.1% (2/22) of the studies. Fatigue monitoring and prevention were addressed in 18.2% (4/22) of the studies. Bone stress-related injuries were examined in 9.1% (2/22) of the studies. Studies focusing on head impacts or conditions related to contact/collision sport history comprised 9.1% (2/22) of the included literature. Other lower extremity pain or injury conditions (e.g., general knee pain, gait asymmetry related to injury history) were investigated in 13.6% (3/22) of studies. Additionally, studies addressed general running-related injury prediction through machine learning applications (4.5%, 1/22), hemophilia-related sports injuries (4.5%, 1/22), and methodological/reliability assessments of gait analysis fundamental for clinical application (4.5%, 1/22).

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Wearable Sensor Technologies Utilized

Sensor Types Employed

IMUs were the most prevalent sensor type, utilized in 59.1% (13/22) of the studies. Accelerometers (either standalone or integrated within IMUs or combined with other sensors) were used in 22.7% (5/22) of studies. sEMG was employed in 22.7% (5/22) of the studies. One study utilized a comprehensive array of integrated sensors, including IMU components, sEMG, and plantar pressure sensors (32). One study used a force sensor (piezoelectric) (23). Breaking down sensor usage by clinical condition: IMUs were used for studies investigating fatigue, bone stress injuries, head impacts, contact/collision sports, hemophilia, and gait analysis reliability. Accelerometers were used, alone or in combination, in studies of fatigue, bone stress, and lower limb injuries, and ACL injuries. Studies focused

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on pain utilized sEMG and, in one case, a piezoelectric force sensor. ACL injury studies employed a range of sensors, including IMUs, accelerometers, sEMG, and a force sensor.

Sensor Placement Strategies

Sensor placement varied across the 22 included studies. The most common placement was on the tibia (either alone or in combination with other locations), used in the majority of studies. Other frequent placements included the foot/ankle region (midfoot, foot, ankle joint, shoelaces), and the lumbar spine/pelvis. Some studies also placed sensors on the sternum, muscles (gastrocnemius, biceps femoris), thighs, waist, and additional ankle locations, and femoral landmarks.

Key Gait Parameters and Biomechanical Insights Derived from Wearable Sensors

This review identified a variety of gait parameters measured by wearable sensors, used to identify biomechanical abnormalities and inform interventions for injury prevention and rehabilitation. Several key gait parameters were frequently reported across the included studies. Tibial shock, a measure correlated with tibial load rates and thus relevant for assessing bone stress injury risk, was often measured using IMUs (37). Vertical ground reaction force (vGRF), frequently quantified in studies of runners, was used to investigate loading patterns, particularly in relation to sacral stress fractures (40) and as a key predictor in running-related injury assessment (32). Muscle activity, assessed using surface electromyography (sEMG), provided insights into muscle activation patterns during gait, with applications in studies examining conditions like anterior knee pain and groin pain (28, 29, 31) and for injury prediction (32). Some studies utilized loading factor trajectories to analyze gait differences, particularly in individuals with ACL deficiency compared to healthy controls (25). Peak acceleration values and vertical ground reaction forces were also quantified with IMUs or force sensors, were used to evaluate stress on the ankle and knee joints, often in the context of knee pain investigations (23, 26). Notably, new contributions include the use of stride interval correlations as sensitive markers of fatigue during training overload (34). Additionally, comprehensive spatio-temporal parameters such as speed, cadence, stride length, step length, and various gait cycle phases were extensively assessed for reliability (33) and integrated into predictive models for running injuries (32). These various gait parameters, captured through IMUs, accelerometers, sEMG, plantar pressure sensors, and force sensors, were instrumental in understanding the nature of gait deviations associated with specific injuries and conditions.

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Main Findings on Clinical Applications

This review's main findings highlight the application of sensor-based gait analysis in identifying biomechanical alterations, supporting injury risk stratification, and informing rehabilitation strategies in athletes.

Wearable sensors demonstrated the ability to detect biomechanical abnormalities, such as altered loading factor trajectories in ACL-deficient individuals (25), abnormal tibial shock in athletes at risk for bone stress injuries (37), and altered muscle activity patterns in those with anterior knee or groin pain (28, 29, 31). Across ten studies addressing injury prevention, seven consistently reported associations between wearable-derived gait parameters, including tibial shock, vertical loading rates, and asymmetry indices, and different types of injuries (25-27, 31, 32, 37, 43). In contrast, three studies found no significant associations between sensor-derived gait parameters and exposure-related outcomes, particularly in those analyzing collision-sport history or activity metrics, suggesting potential sport-specific variability in sensor sensitivity (24, 35, 38). These discrepancies were most evident in investigations using load-rate or exposure-based proxies for injury risk.

Changes in running stride-interval correlations (34) were also identified as sensitive markers of fatigue accumulated during overload training, providing a field-based method for monitoring training tolerance. Four studies evaluated performance-related applications such as fatigue monitoring and return-to-play decisions. Two studies demonstrated clear associations between stride-interval correlations, fatigue, and performance decline, whereas two others yielded inconclusive or confounded findings, with speed and stride frequency accounting for much of the variance in mechanical changes and masking stage effects until corrected, revealing only weak associations between biomechanical metrics and perceived exertion (36, 39).

Furthermore, advanced machine learning approaches using wearable sensor data reported high classification performance for running-related injury prediction, identifying key biomechanical factors such as ground reaction force, foot pressure, and stride length as important predictors in single-cohort analyses (32). The ability to detect these deviations has implications for both injury prediction and prevention. For example, measurements of tibial shock and vertical ground reaction force were shown to be relevant for assessing bone stress injury risk (37, 40) and for broader running injury prediction models (32).

Among eight rehabilitation-focused studies, six demonstrated consistent utility of wearable sensors in monitoring recovery after ACL reconstruction or hip and groin pathology.

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Conversely, two studies reported non-significant or condition-dependent muscle activation findings, with limited between-group differences despite EMG pattern variations (28, 30). Such variability appears to stem from differences in sensor placement, task selection, and outcome definitions. Wearable sensor-based gait analysis nonetheless provided valuable data for monitoring rehabilitation progress, allowing for objective assessment of movement patterns and informed adjustments to treatment protocols.

The reliability of wearable IMUs for measuring a wide range of gait parameters, including speed, cadence, and various gait cycle phases, was also confirmed across sessions (33), underscoring their foundational utility for consistent longitudinal monitoring in clinical and training settings. In summary, while methodological quality varied, the majority of studies provided consistent evidence supporting the clinical value of wearable sensors in both injury prevention and rehabilitation, with some conflicting findings highlighting the need for more standardized protocols. These findings should be interpreted with caution, because small samples and incomplete methodological reporting may limit generalizability and reproducibility.

Discussion

Overview of Key Findings and Significance

This systematic review comprehensively synthesizes current research on the diverse benefits, inherent limitations, and crucial future directions of wearable gait analysis in sports medicine and athletic training.

Wearable sensor technologies, such as IMUs and sEMG, offer a significant advantage over traditional laboratory-based gait analysis by providing dynamic, real-time assessment of athletes' movements in naturalistic training and competition settings. This capability is crucial, as this review found that wearable sensors can identify specific biomechanical abnormalities, including altered loading factor trajectories in ACL-deficient individuals (25), abnormal tibial shock in those at risk for bone stress injuries (37), and altered muscle activity patterns in individuals with anterior knee pain or groin pain (28, 29, 31). Beyond these, recent advancements demonstrate their utility in monitoring fatigue through stride interval correlations (34) and predicting running-related injuries by identifying key biomechanical factors such as ground reaction force, foot pressure, and stride length through machine learning models (32). The robust reliability of these wearable IMUs for various gait parameters further underscores their practical applicability (33). This capacity to detect subtle gait deviations and provide comprehensive insights demonstrates the significant

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potential of wearable sensors for enhancing both preventative and rehabilitative care in sports medicine.

We assessed overall confidence in key findings using a structured qualitative grading approach consistent with the strength-of-evidence framework applied in the Results section. Confidence ratings reflected study quantity, methodological consistency, and design rigor. Moderate confidence was assigned for tibial accelerometer–based detection of bone stress injury risk (5 studies, consistent outcomes, moderate-quality cross-sectional designs). Low confidence was assigned for IMU-based ACL rehabilitation monitoring (4 studies, heterogeneous designs, limited sample sizes, variable outcome metrics). Moderate confidence was assigned for stride-interval correlation as a fatigue marker (2 studies, robust methodology but limited replication). These levels reflect the synthesized evidence base rather than formal GRADE scoring.

Comparative Clinical Utility of Different Sensor Technologies

Sensor selection should follow the clinical question. IMUs and accelerometers, the most prevalent modalities, quantify spatiotemporal parameters, impact loads such as tibial shock relevant to bone stress risk (37, 40), and movement kinematics indicative of fatigue or injury (32, 34). In contrast, sEMG captures muscle activation timing and intensity, which is useful for investigating neuromuscular control in knee or groin pain (28, 29, 31) and for monitoring muscle fatigue. Plantar pressure and discrete force sensors, though less frequently used, uniquely measure foot-ground interaction forces and pressure distributions, which inform foot biomechanics. Multimodal integration (32) can provide a more complete biomechanical profile, while practical factors such as usability and data management also influence device choice.

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Building on these comparisons, clinicians can apply concise selection and placement guidance. IMUs are suitable for longitudinal monitoring and spatiotemporal assessment (33). Accelerometers are well suited to detecting tibial shock and vertical loading rates in runners at risk for bone stress injuries (37, 40). sEMG is most informative when evaluating neuromuscular activation in athletes with anterior knee or groin pain (28, 29, 31). Placement consistency is essential: tibial-mounted accelerometers optimize shock detection (37, 40), waist or trunk-mounted IMUs provide robust stride length and cadence data (33), and surface electrodes over hip and thigh musculature best capture activation patterns (28, 29, 31). These considerations can guide sensor selection and placement for specific clinical scenarios.

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Clinical Applications in Injury Prevention

A key application of wearable sensor-based gait analysis lies in injury prevention. This review found that wearable sensors can identify individuals at increased risk for specific injuries. For example, studies using IMUs demonstrated the potential to identify runners at higher risk of bone stress injuries by detecting abnormal tibial shock (37) and altered vertical ground reaction forces (40). Furthermore, the ability to detect gait asymmetries, as shown in studies of runners (43) and individuals with ACL deficiency (25), highlights the potential for early intervention to correct biomechanical imbalances before they lead to injury. Studies investigating muscle activity patterns using sEMG also provided valuable insights into the mechanisms underlying hip and groin pain (28, 29), suggesting potential targets for preventative interventions. Most notably, recent advances demonstrate the powerful capability of machine learning algorithms, coupled with multi-modal wearable sensor data (including IMUs, sEMG, and plantar pressure sensors), to accurately predict running-related injuries by identifying critical factors such as ground reaction force, foot pressure, and stride length (32). This predictive capacity, along with the ability to monitor fatigue progression through stride interval correlations (34), offers a significant step forward in proactive injury risk assessment and training load management in real-world settings. The confirmed reliability of IMU-based gait parameters also underpins the robustness required for consistent, long-term injury surveillance (33). Overall, the evidence indicates that wearable sensor technologies hold substantial promise for enhancing preventive care in sports medicine, enabling more data-driven and proactive athlete management.

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Clinical Applications in Rehabilitation

Wearable sensor-based gait analysis also plays a crucial role in rehabilitation. The objective data provided by these sensors allow for precise monitoring of recovery progress and inform evidence-based decisions regarding return-to-play timelines. This review identified studies demonstrating the utility of wearable sensors in assessing gait deviations in individuals recovering from ACL reconstruction (23, 42), and understanding muscle activity changes during recovery from anterior knee pain (31). Furthermore, the confirmed reliability of IMU-based systems for measuring a wide range of spatio-temporal gait parameters (33) support their foundational utility for consistent longitudinal monitoring throughout the rehabilitation process. These findings suggest that wearable sensors can be used to tailor rehabilitation programs to address specific biomechanical deficits, potentially leading to improved outcomes and a safer return to sport. Specifically, the integration of machine learning with wearable data holds promises for objectively assessing readiness for return-to-sport by

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identifying critical biomechanical patterns indicative of persistent injury risk (32). Moreover, monitoring subtle markers of fatigue, such as changes in stride interval correlations during graded exercise, can inform safe and progressive loading strategies during rehabilitation (34). Overall, these results underscore the immense potential of wearable sensor technologies to enhance rehabilitative care in sports medicine, facilitating highly individualized and data-driven recovery pathways.

Clinical Implementation Recommendations

For injury prevention, the evidence is moderate, based on cross-sectional exertion protocols and observational field studies. Some studies in this review reported within-athlete increases in tibial shock and vertical loading rates on the order of 5–7 percent during exertion, but there are no universal asymmetry cutoffs, and such changes should be interpreted longitudinally within athletes rather than as fixed thresholds (37, 42, 43). Recommended configurations include tibial-mounted IMUs (≥1000 Hz, ±16 g) for load detection, shoe-mounted IMUs at 1000 Hz for contact-time asymmetry, and ViMove tibial sensors coupled with TekScan pressure mats for alignment and loading assessment.

For fatigue and performance monitoring, the evidence is moderate, derived from longitudinal overload or taper studies and outdoor observational data. Observed changes typically include a decrease in stride-interval complexity (DFA α1) during overload with recovery during taper, while speed and stride frequency account for roughly 20–30 percent of mechanical variance, requiring correction to isolate fatigue effects (34, 36, 39). Appropriate setups employ tibial or L3–L5 IMUs sampling at 100–201 Hz for stride metrics or multi-IMU arrays at 240 Hz with GNSS for outdoor gait modeling (34, 36, 41).

For rehabilitation, the evidence is moderate, drawn from cross-sectional sEMG profiling and repeated-measures body-weight-support running protocols. No universal numeric cutoffs exist; instead, pattern normalization and within-athlete change should guide progression (28, 30). Recommended systems include wireless sEMG sampled near 2000 Hz with SENIAM placement over hip and thigh musculature and treadmill protocols with graded body-weight support using synchronized IMU or sEMG signals for gait-event timing.

For return-to-play readiness and longitudinal monitoring, the evidence is moderate, based on test–retest analyses. Established benchmarks include MDC_{95} < 7 percent and inter-session ICCs between 0.68 and 0.93 for spatiotemporal gait parameters (33). These metrics are typically derived from lumbar IMUs operating near 100 Hz for cadence, stride length, and double-support assessments.

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Across these domains, minimal clinically important differences for tibial shock, loading rate, stride-interval complexity, and gait symmetry have not been formally standardized. The reported percentage changes represent synthesized estimates from available studies. In practice, clinicians should compare observed changes with reliability-based indices such as ICC, SEM, and MDC for their own device and protocol, so that observed differences are more likely to represent meaningful biomechanical adaptation rather than measurement variability.

Methodological Considerations and Limitations of Included Studies

Despite their benefits, several significant challenges must be systematically addressed to fully realize the clinical potential of wearable sensors in sports medicine. The methodological quality assessment of the included studies revealed key limitations. Variability in reporting methodology, including sensor placement, data collection procedures, sensor types, analytic methods, and outcome measures, was a pervasive concern. This heterogeneity, also noted in recent systematic reviews (16), impedes direct comparison of findings and restricts synthesis of a cohesive evidence base. Quantitative differences across methodological settings further illustrate this variability. For example, studies using 50 Hz sampling may underestimate peak tibial acceleration by approximately 10–15% compared with 200 Hz recordings, leading to underestimation of impact loads (37, 40). Similarly, low-pass filter cutoffs between 50 and 100 Hz can modify calculated tibial shock amplitudes by 20–30%, depending on the dominant frequency content of running impacts (37, 41). These values represent approximate synthesized estimates derived from the collective range reported in included studies.

Sensor placement also contributes: waist- or lumbar-mounted IMUs record axial acceleration magnitudes approximately 30–40% lower than tibial-mounted devices (33, 43), which may reduce sensitivity for detecting bone-stress risk.

For instance, differences in IMU placement or analytic approaches to defining gait events contributed to inconsistent associations between gait parameters and injury risk, thereby hindering the development of robust, evidence-based guidelines. Many non-comparative studies were also limited by small sample sizes and the absence of control groups. Additional weaknesses included the lack of prospective sample size calculations, which may undermine statistical power, and significant loss to follow-up in several studies, increasing the risk of attrition bias. These limitations underscore the critical need for future research to adopt more rigorous designs and standardized protocols.

Beyond study design issues, critical considerations for clinical translation include ensuring data accuracy and inter-device reliability. If clinicians encounter notable discrepancies in

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metrics such as ground contact time or impact loading across devices, or if outputs are overly complex without clear interpretative guidance, confidence in using these tools for key decisions such as return-to-play clearance may be diminished. Effectively translating complex biomechanical data into intuitive, actionable insights while safeguarding data privacy also remains vital for broader adoption.

Importantly, the heterogeneity observed across studies has direct clinical implications. For example, tibial-mounted accelerometers yield substantially higher tibial shock estimates than waist-mounted IMUs, which could lead to different interpretations of bone stress injury risk. Similarly, proprietary machine learning models versus open-source algorithms can produce divergent estimates of stride variability or fatigue detection, potentially altering return-to-play recommendations. Such inconsistencies highlight the need for standardized sensor placement protocols and transparent algorithm reporting before wearable sensor data can be reliably integrated into clinical practice.

Beyond methodological limitations, broader implementation of wearable sensor technologies faces additional challenges. Regulatory pathways and the absence of universally accepted standards for sensor validation, calibration, and data interoperability may delay clinical adoption. Practical implementation barriers also merit consideration. The cost of wearable sensor systems varies widely, with research-grade or clinical setups (e.g., Xsens, Noraxon, Delsys) typically priced around $5,000–$15,000, while consumer-grade devices (e.g., RunScribe, Shimmer) range from $200–$800, based on published manufacturer pricing and market estimates as of 2025. Operating these systems requires structured training of approximately 8–12 hours, generally covering calibration, data acquisition, and signal interpretation, based on vendor documentation and authors' professional experience. From a regulatory perspective, most commercially available systems are not FDA-cleared or CE-marked for diagnostic use, remaining designated for research or performance monitoring. At present, no standardized certification programs exist for clinicians or sports scientists in wearable gait analysis, underscoring the need for formalized training and credentialing frameworks before broad clinical integration. Cost-effectiveness is another concern, as high device costs and ongoing maintenance can limit access, particularly in resource-constrained settings. Furthermore, successful integration into sports medicine practice requires adequate training for clinicians, ensuring that complex biomechanical data are interpreted consistently and translated into actionable clinical decisions. Addressing these regulatory, economic, and educational factors will be critical for sustainable adoption of wearable sensor–based gait analysis in clinical practice.

To optimize clinical decision-making, wearable sensor data should be interpreted in conjunction with traditional assessments rather than in isolation. Integration with physical

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examination findings, strength or range-of-motion testing, and patient-reported outcome measures enhances contextual understanding and ensures that biomechanical metrics align with functional recovery and perceived readiness. This multimodal approach supports balanced, evidence-informed decisions across injury prevention, rehabilitation, and return-to-play management.

Limitations of the Systematic Review

This systematic review, while comprehensive in scope, has several inherent limitations that should be considered. Firstly, due to the substantial heterogeneity in study designs, participant populations, sensor types, gait parameters, and outcome measures, a quantitative meta-analysis was not feasible. Consequently, our findings are based on a narrative synthesis, which, while appropriate for summarizing diverse evidence, may be subject to reviewer interpretation. Secondly, despite a thorough search of major databases, there remains a potential for publication bias, as studies with null or negative findings may be underrepresented in the published literature. Finally, our review was limited to peer-reviewed journal articles, and the exclusion of grey literature such as conference proceedings, dissertations, and unpublished reports. We cannot determine whether grey literature exclusion introduced bias, as we did not systematically catalog excluded grey literature. However, restriction to peer-reviewed publications enhances methodological transparency and ensures consistent application of inclusion criteria. No updated search was performed after May 2025, and the evidence base reflects the state of knowledge as of that date.

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Future Directions

To advance the clinical utility of wearable sensor-based gait analysis, future research should focus on several key areas. Central to this is leveraging technological advancements to overcome identified limitations. For instance, the development of next-generation multi-modal sensors (integrating IMUs, sEMG, and even biochemical markers) with enhanced on-body processing capabilities can improve data fidelity and reduce noise, directly tackling issues of data accuracy. Advancements in artificial intelligence (AI) and machine learning (ML) are pivotal for developing algorithms that can automatically identify relevant gait events, extract meaningful features from large datasets, and adapt to individual variations, thereby addressing the current lack of standardized data processing and interpretation. These AI/ML models can also facilitate the creation of robust predictive analytics for injury risk that account for complex, non-linear interactions.

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Beyond sensor and algorithm development, standardized protocols for data collection (including sensor placement and calibration) and transparent reporting are essential to enhance comparability across studies. Longitudinal studies are needed to establish the long-term predictive validity of wearable sensor data for injury prevention and to track rehabilitation progress effectively. Finally, exploring the integration of wearable sensor data with other clinical information sources (e.g., medical history, imaging, training load, psychological assessments) within comprehensive athlete management platforms will provide a more holistic understanding of injury risk and optimize personalized interventions, truly ushering in an era of data-driven sports medicine.

Conclusion

Wearable sensor–based gait analysis shows emerging evidence of providing objective biomechanical insights that may support injury risk assessment and rehabilitation monitoring in athletic populations. Current evidence suggests that simple, strategically placed sensors may provide clinically useful information on loading patterns and spatiotemporal control, complementing standard clinical evaluations and decision-making in sports medicine. Emerging evidence suggests that tibial accelerometers may be useful for monitoring impact loads related to bone stress injuries, while trunk-mounted IMUs show promise for capturing spatiotemporal parameters that inform rehabilitation progress. However, these recommendations are derived predominantly from moderate-quality observational studies, and clinicians should apply appropriate judgment and pursue institutional validation before clinical implementation.

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Readers should note that most included evidence demonstrated moderate methodological quality (median MINORS 17/24 for comparative designs and 11/16 for non-comparative designs), with common limitations including small sample sizes, heterogeneous sensor protocols, lack of standardized algorithms, and a predominance of cross-sectional designs. These methodological constraints should be considered when interpreting the generalizability of the findings.

Institutions aiming to integrate wearable sensor–based gait analysis into clinical practice should: (1) establish standardized protocols for sensor placement and data collection; (2) validate device output against criterion standards in their specific athlete populations; (3) develop explicit decision-making algorithms linking gait parameter thresholds to clinical actions; and (4) ensure adequate clinician training in data interpretation. Future studies should employ adequately powered samples with standardized sensor protocols to improve comparability, use prospective longitudinal designs to validate injury prediction models and

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define clinically meaningful gait thresholds, and establish reliability and validity of sensor–algorithm combinations across diverse athletic populations. Continued standardization and high-quality prospective research will be essential to advance these technologies from emerging tools to validated components of evidence-based sports medicine practice.

Declarations

Ethics approval and consent to participate: This study is a systematic review of previously published literature and did not involve human participants or animal subjects directly.

Consent for publication: Not Applicable.

Availability of data and materials: The data supporting the findings of this study are available within the cited published articles.

Competing interests: The authors declare that they have no competing interests.

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